

DIGITAL LOGIC DESIGN

(CE_118)

CHAPTER 3:

Register Transfer Specification & Design (part_3)

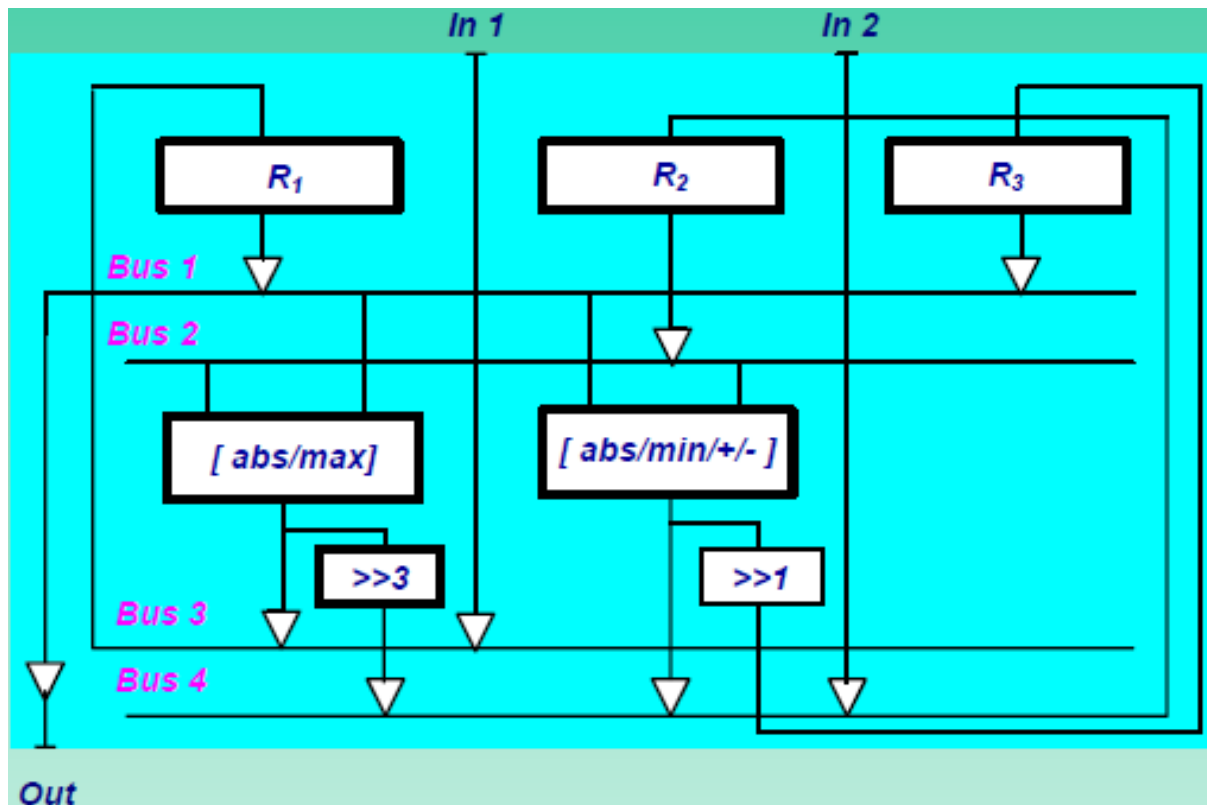
Outline

- **Register Transfer (RT) design:**
 - FSMD model
 - RT specification with
 - Static-action tables
 - ASM charts
- **Procedure for synthesis from RT specification**
- **Design Optimization through**
 - Register sharing
 - Functional unit sharing
 - Bus sharing
 - Unit chaining and Multiclocking
- **Design Pipelining**
 - Unit pipelining
 - Control pipelining
 - Datapath pipelining
- **Scheduling of flowcharts**

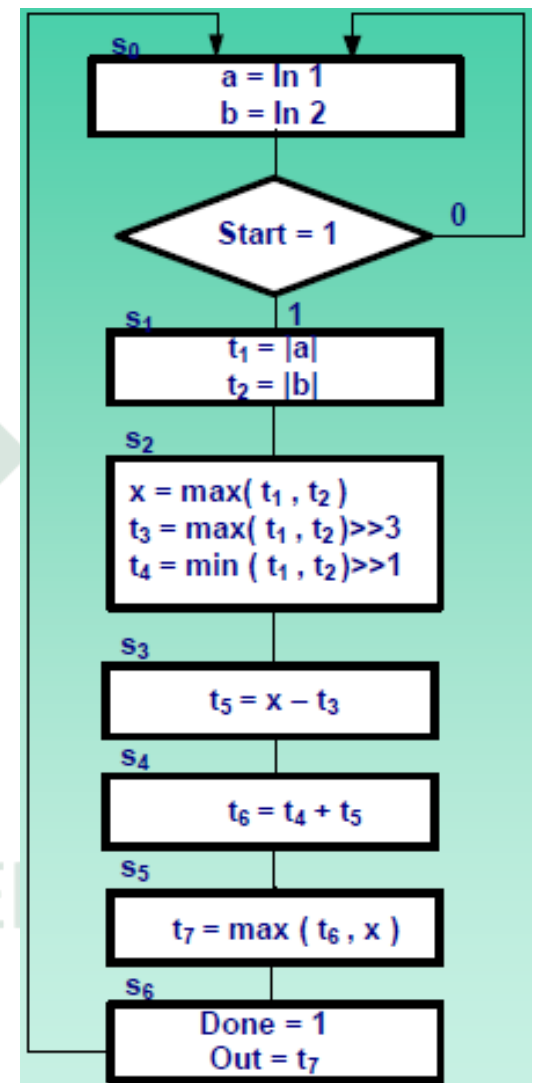
Chaining and multicycling

- Chaining allows serial execution of two or more operations in each state
- Chaining reduces number of states and increases performance
- Multicycling allows one operation to be executed over two or more clock cycles
- Multicycling reduces size of functional units
- Chaining and multicycling are used on noncritical paths to improve resource utilization and performance

SRA datapath with chained units

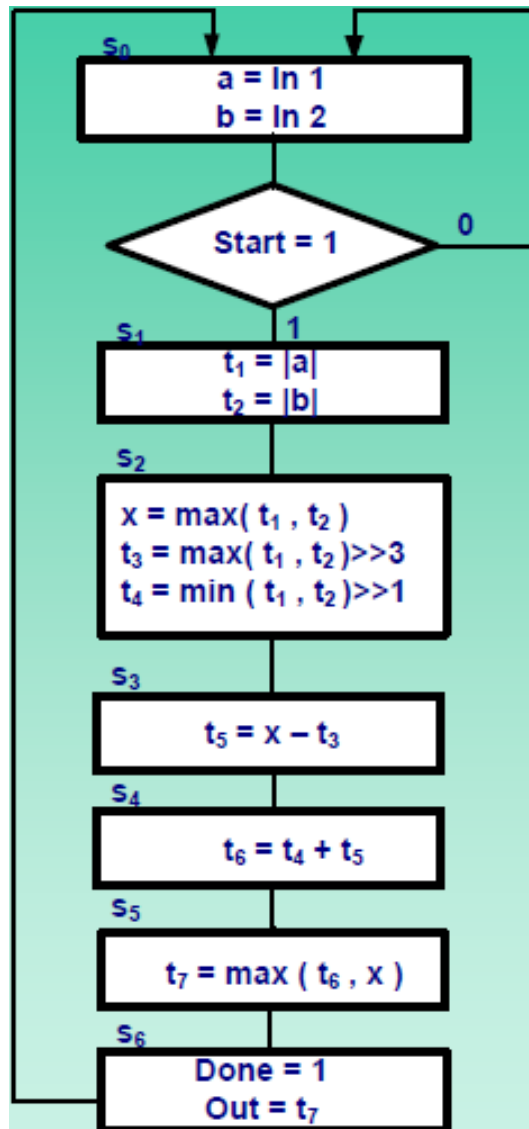


(b) Datapath schematic

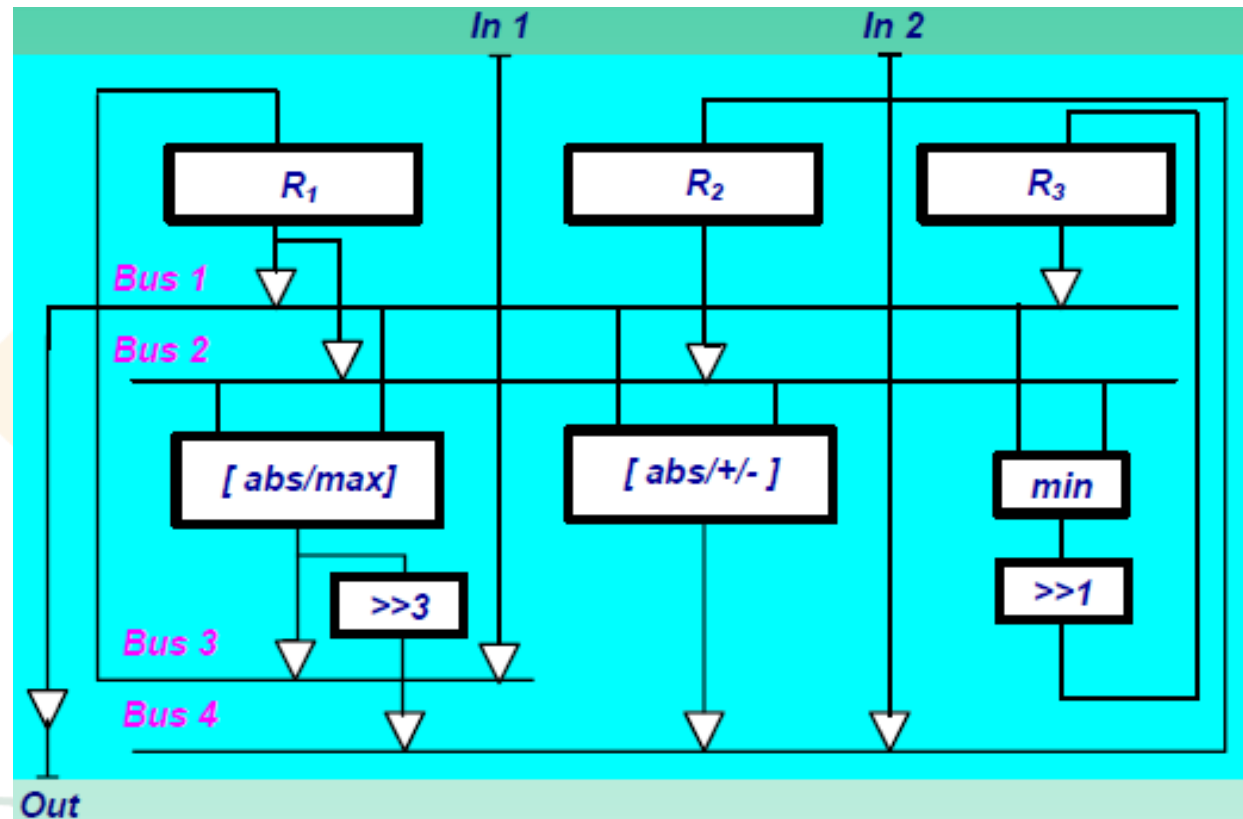


(a) ASM Chart

SRA datapath with multicycle units



(a) ASM Chart

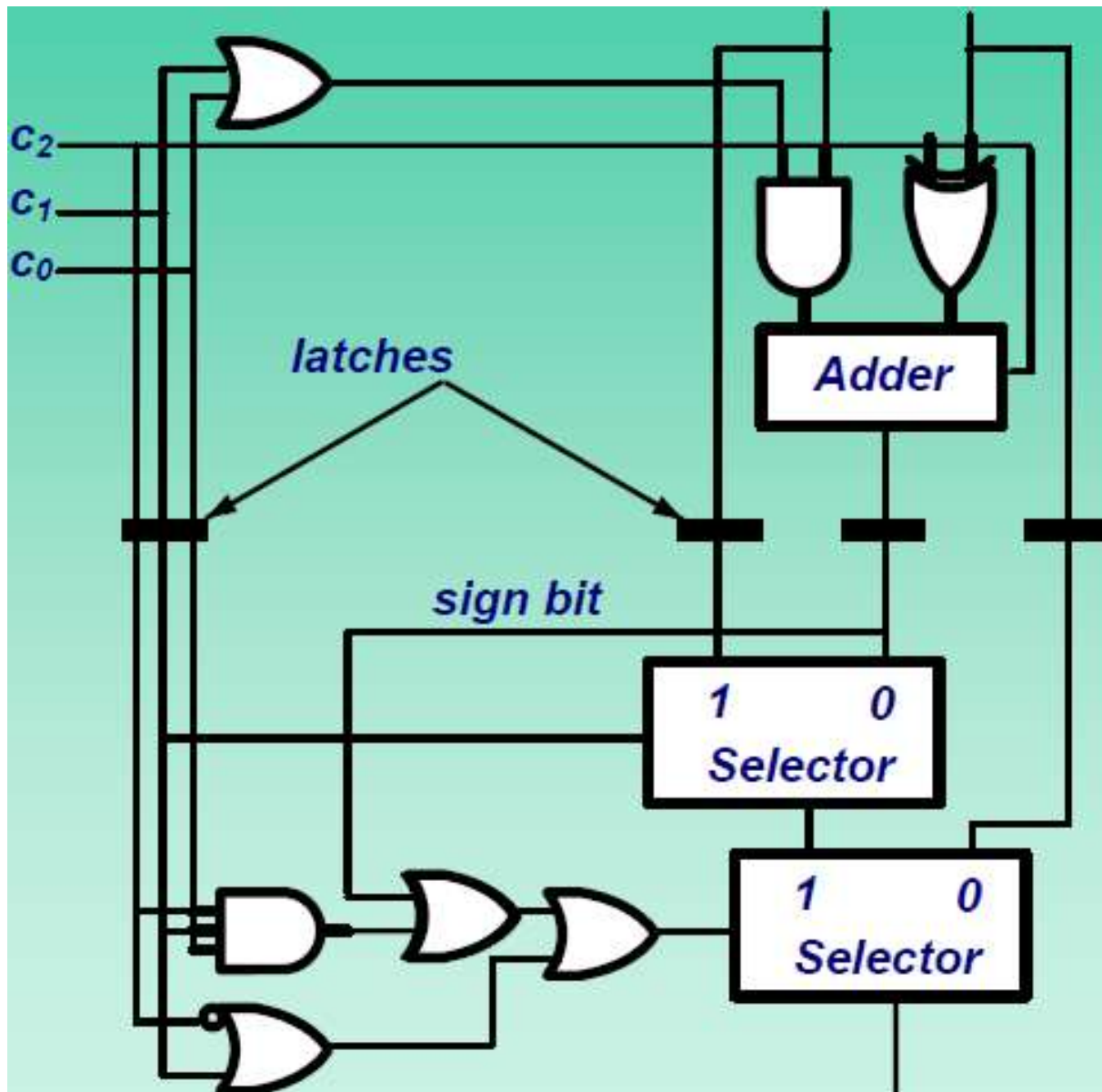


(b) Datapath schematic

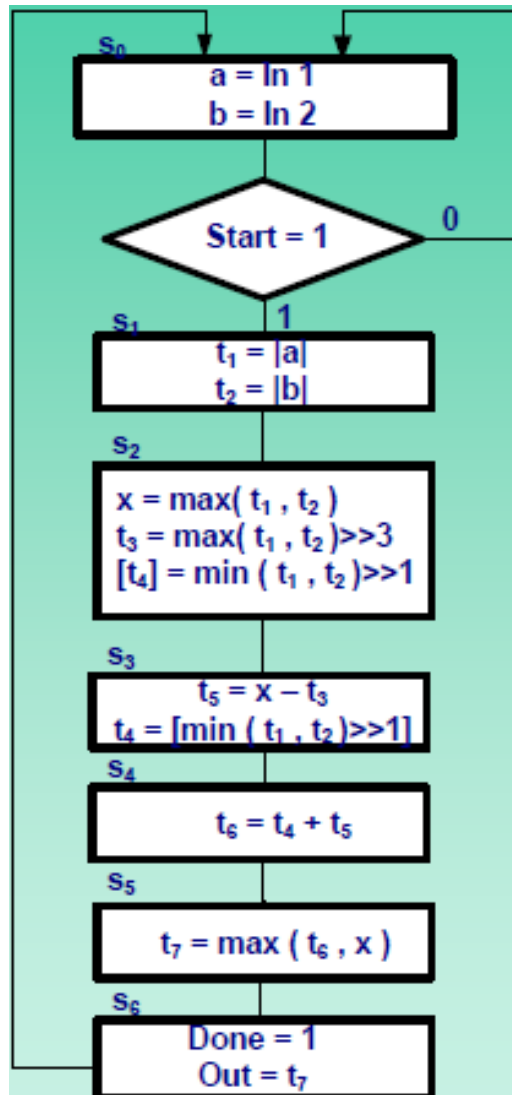
Pipelining

- Pipelining improves performance at a very small additional cost
- Pipelining divides resources into stages and uses all stage concurrently for different data (assembly line principle)
- Pipelining principles works on several levels:
 - **(a) Units pipelining**
 - **(b) Control pipelining**
 - **(c) Datapath pipelining**

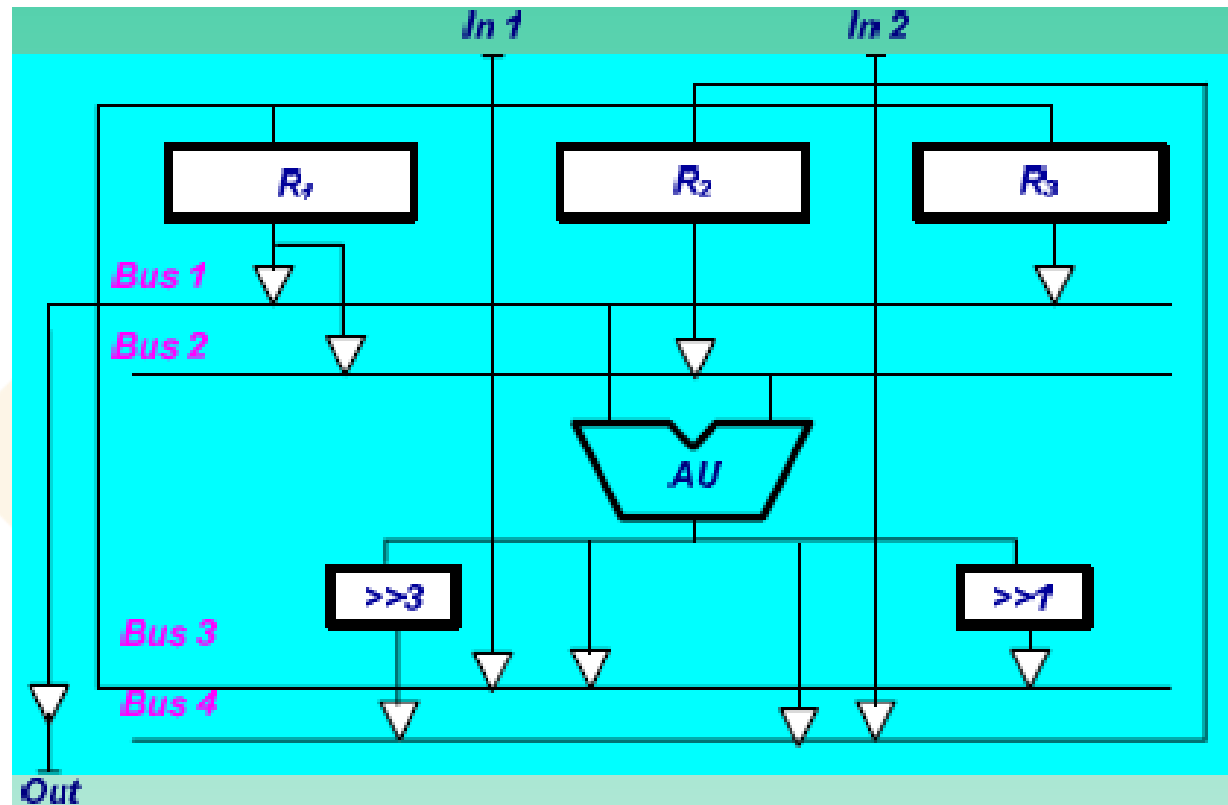
Pipelined arithmetic unit



SRA datapath with single AU

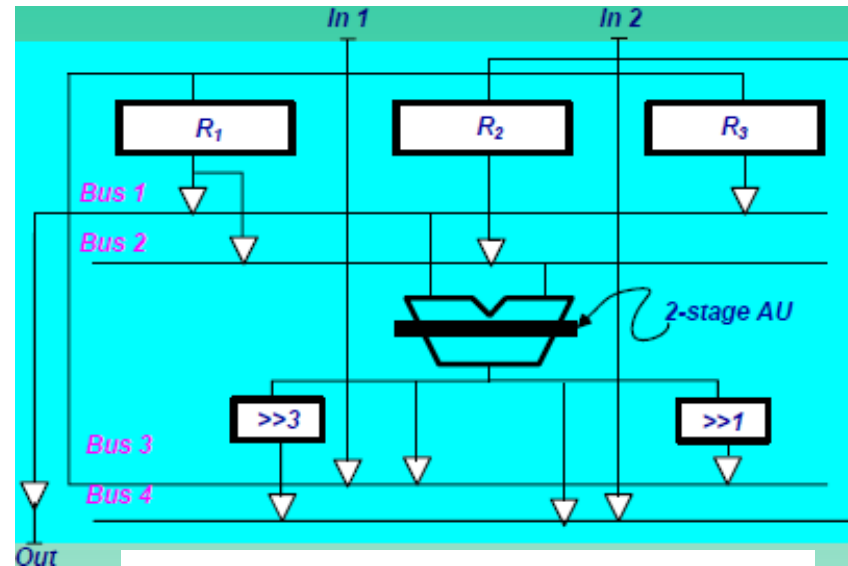


(a) ASM Chart



(b) Datapath schematic

Datapath with pipelined functional unit

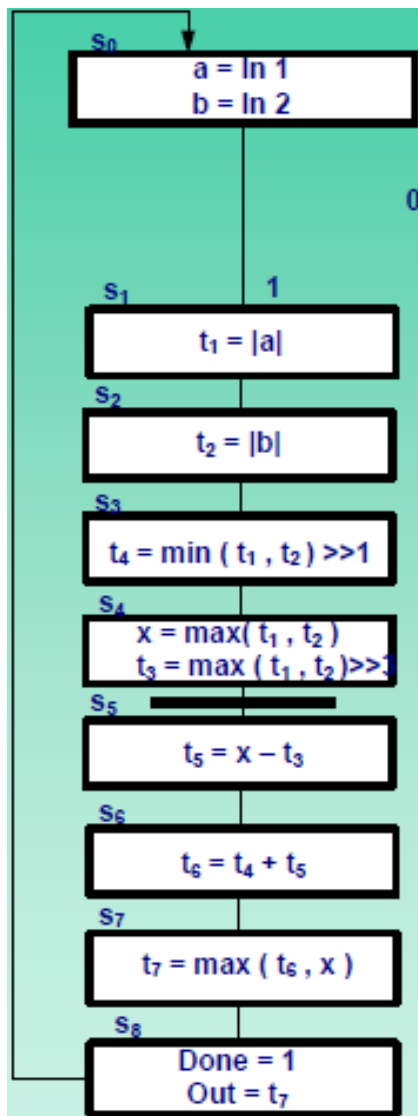


(a) Datapath with pipelined AU

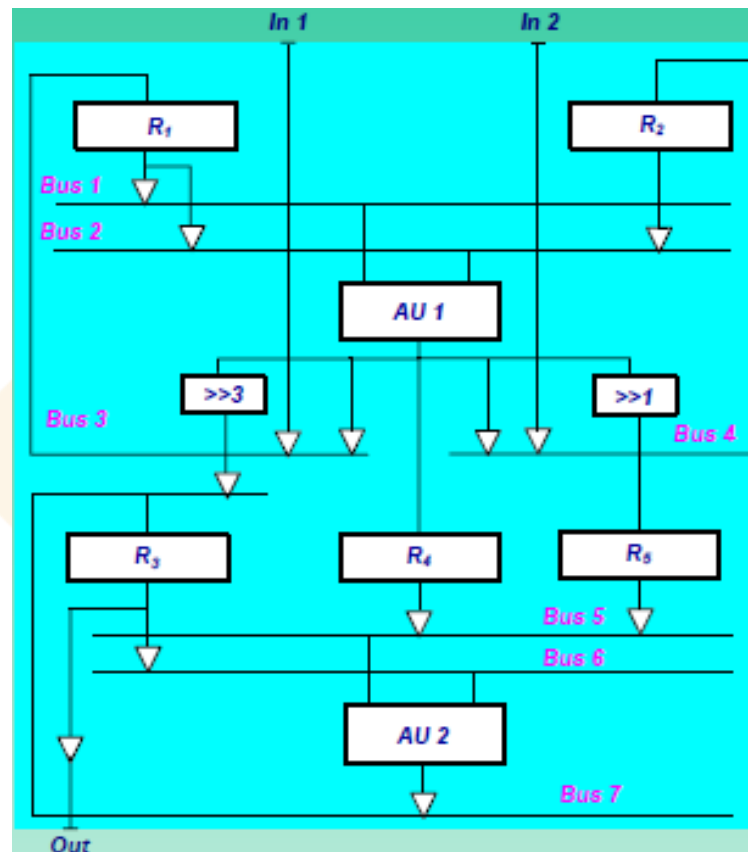
	s_0	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8	s_9	s_{10}	s_{11}	s_{12}
Read R_1		a			t_1	t_1	x				x		t_7
Read R_2			b		t_2	t_2	t_3		t_5		t_6		
Read R_3									t_4				
AU stage 1		a	b		max	min	-		+		max		
AU stage 2			a	b		max	min	-		+		max	
shifters						>>3	>>1						
Write R_1	a		t_1			x						t_7	
Write R_2	b			t_2		t_3		t_5		t_6			
Write R_3							t_4						
Output													t_7

(b) Timing diagram

Datapath pipelining



(a) ASM Chart



(b) Pipelined datapath

R1 = [a, t1]

R2 = [b, t2]

AU1 = [abs/min/max]

R3 = [t3, t5, t6, t7]

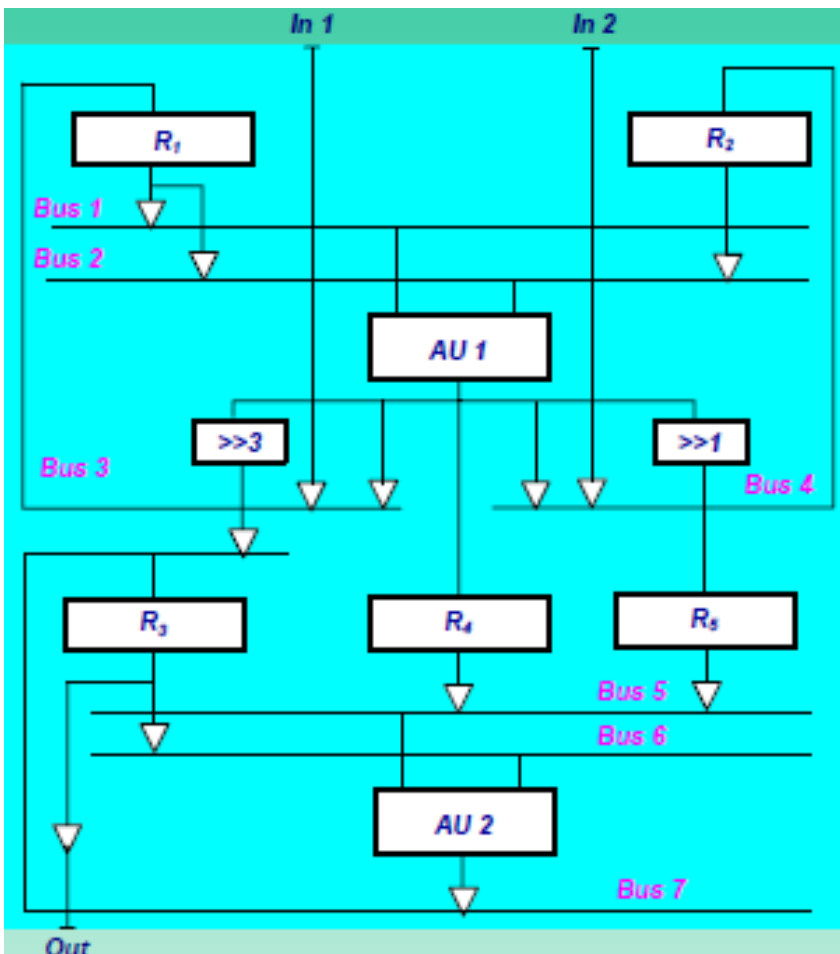
R4 = [x]

R5 = [t4]

AU2 = [+/-/max]

(c) Register and functional unit assignment

Datapath pipelining

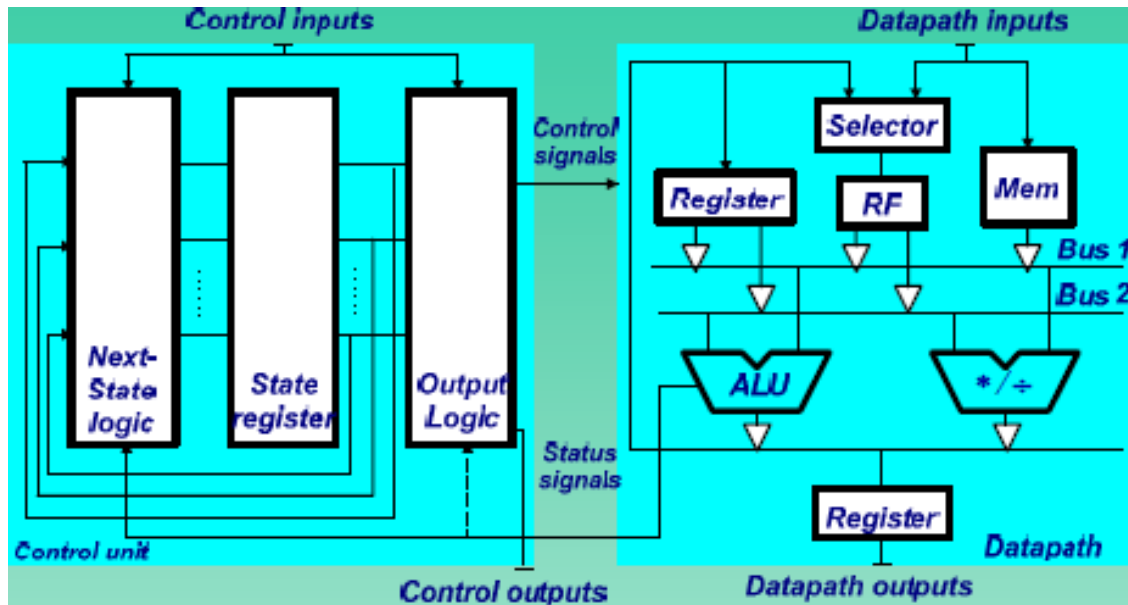


(b) Pipelined datapath

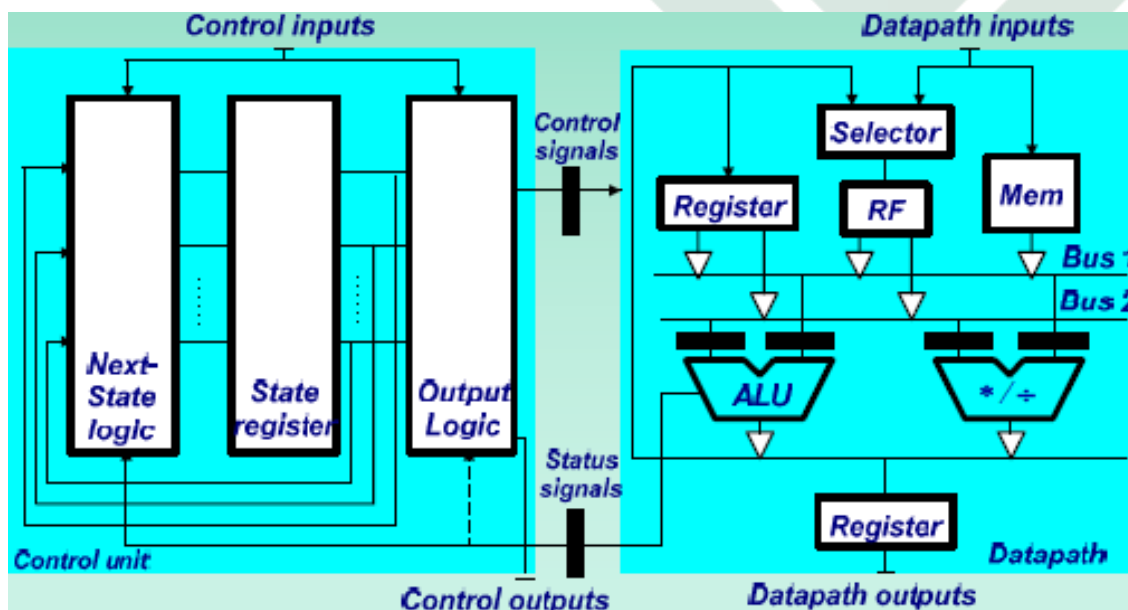


(d) Timing diagram

Pipelined FSMD implementation

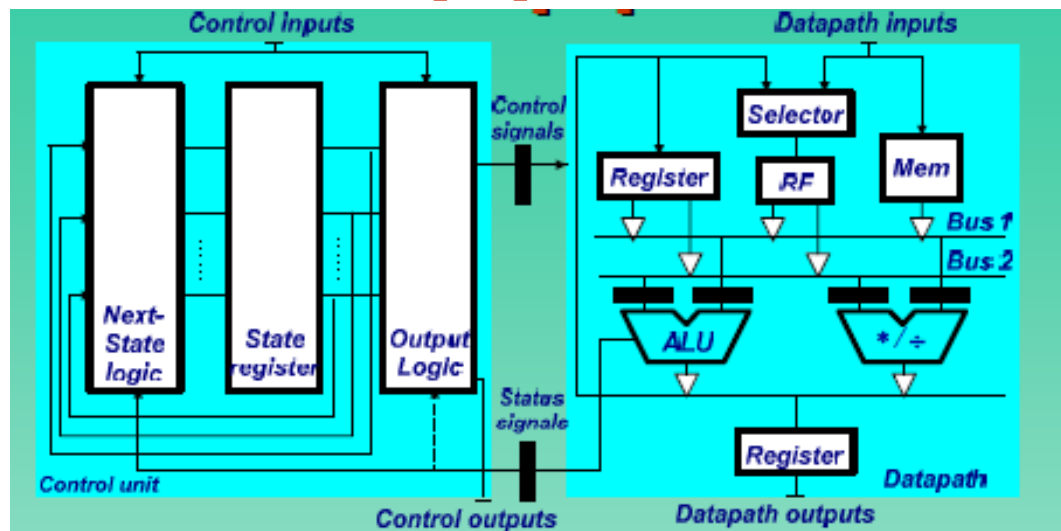


(a) Standard FSMD implementation

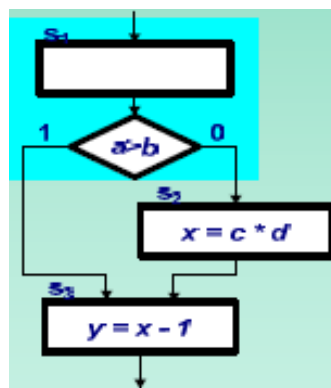


(b) FSMD implementation with control and datapath pipelining

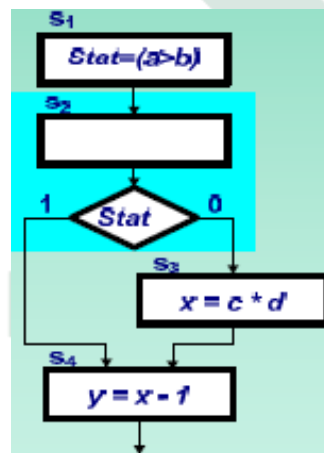
ASM charts for pipelined FSMDs



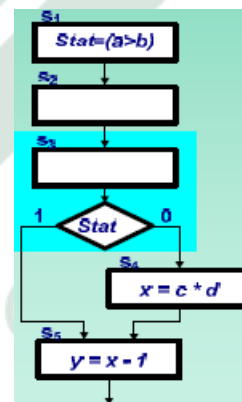
(b) FSMD implementation with control and datapath pipelining



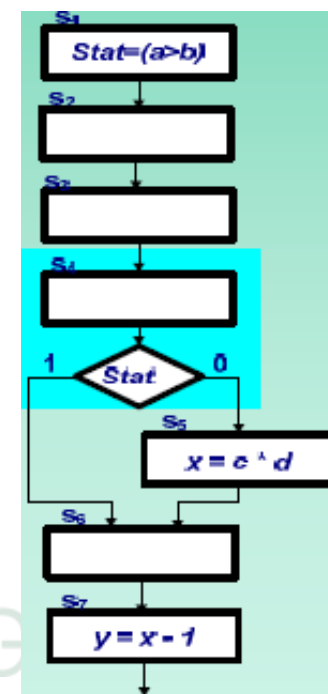
(a) ASM chart



(b) ASM chart for control pipeline with status register



(c) ASM chart for control pipeline with status register and control registers

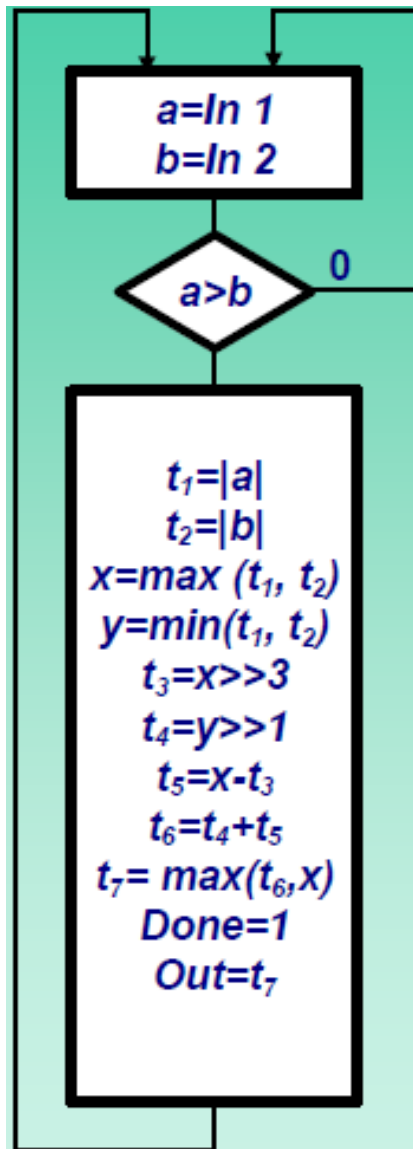


(d) ASM chart for control and datapath pipeline

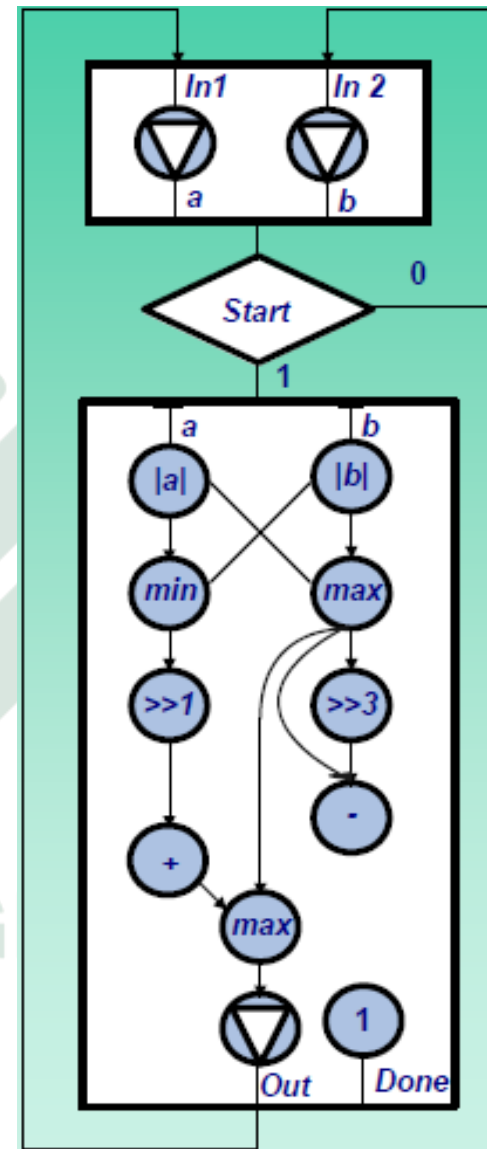
Scheduling

- RT description such as ASM chart specifies data operations in each state.
- Flowcharts or programming languages do not have states, but only specify order in which operations are executed.
- Scheduling transforms flowcharts or programs with RT descriptions.
- Two types of scheduling:
 - (a) resource constrained (resource given, minimize time)
 - (b) time constrained (time given, minimize resources)

Control/dataflow graph for SRA

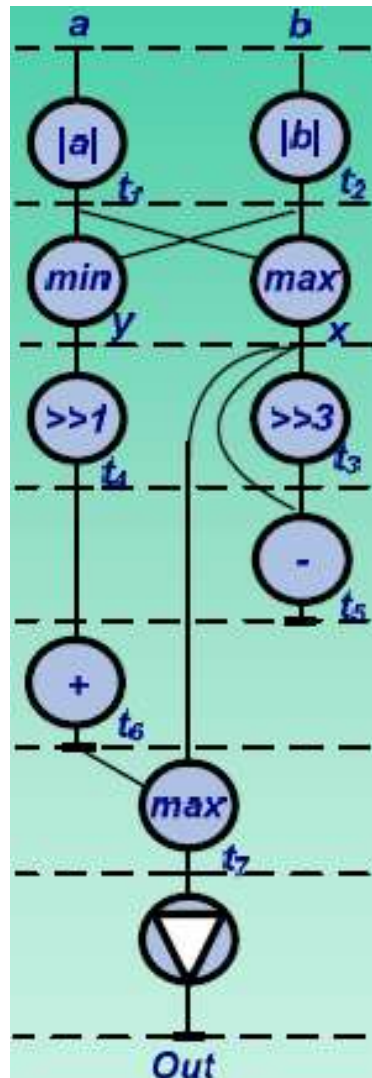


(a) Flowchart

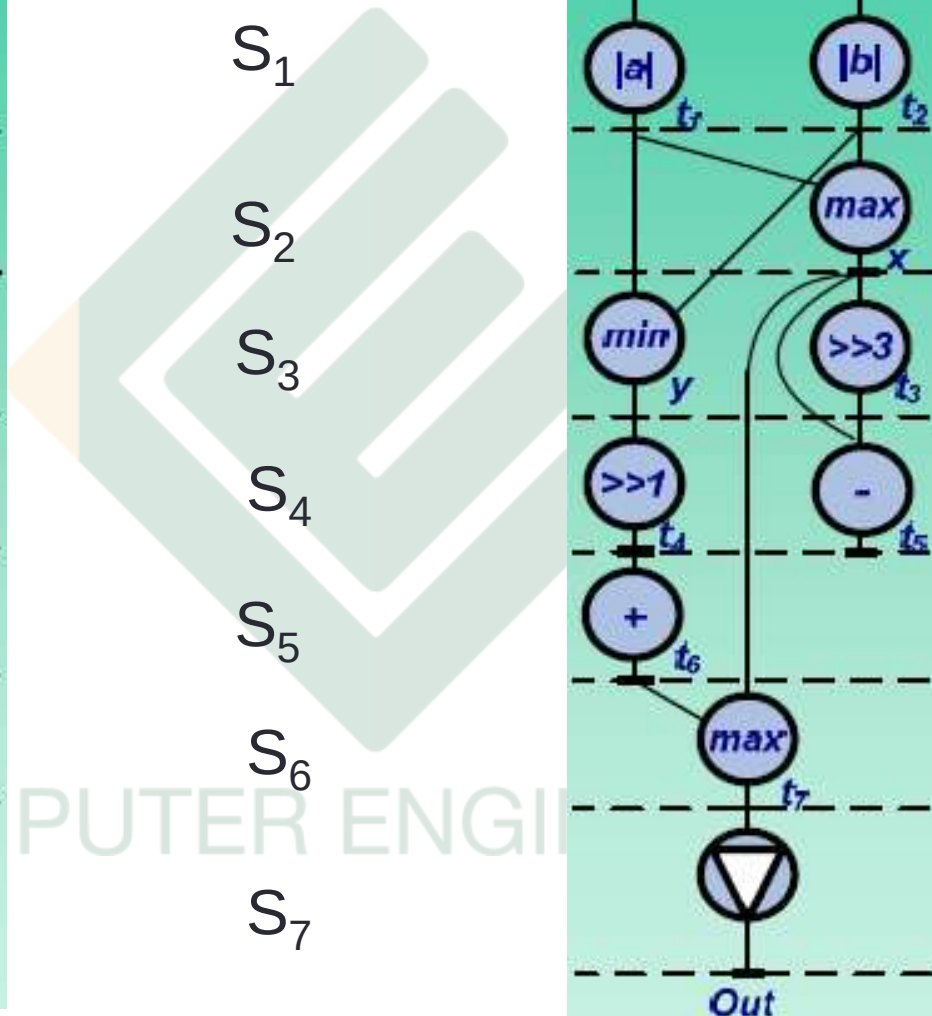


(b) Control/Data flow graph

Basic schedules

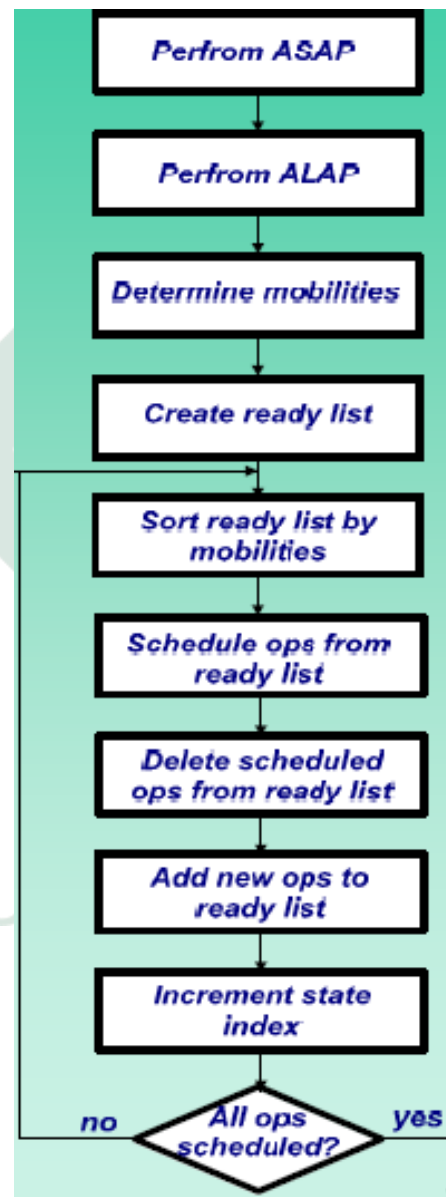


(a) ASAP schedule

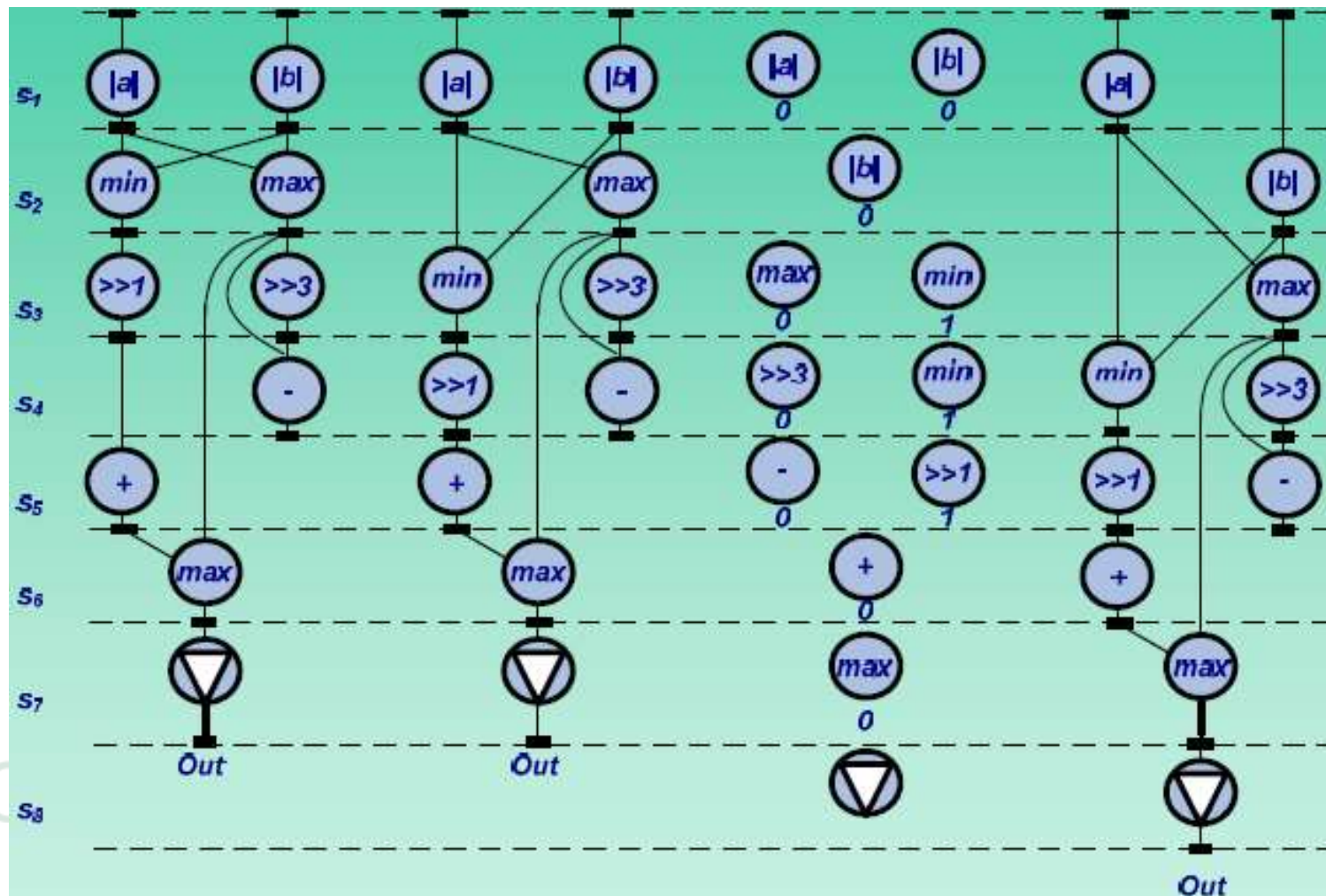
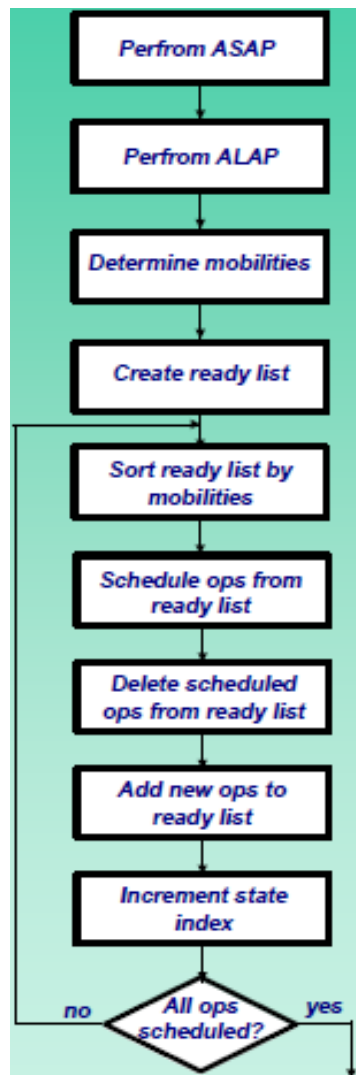


(b) ALAP schedule

List scheduling algorithm



Resource-constrained scheduling

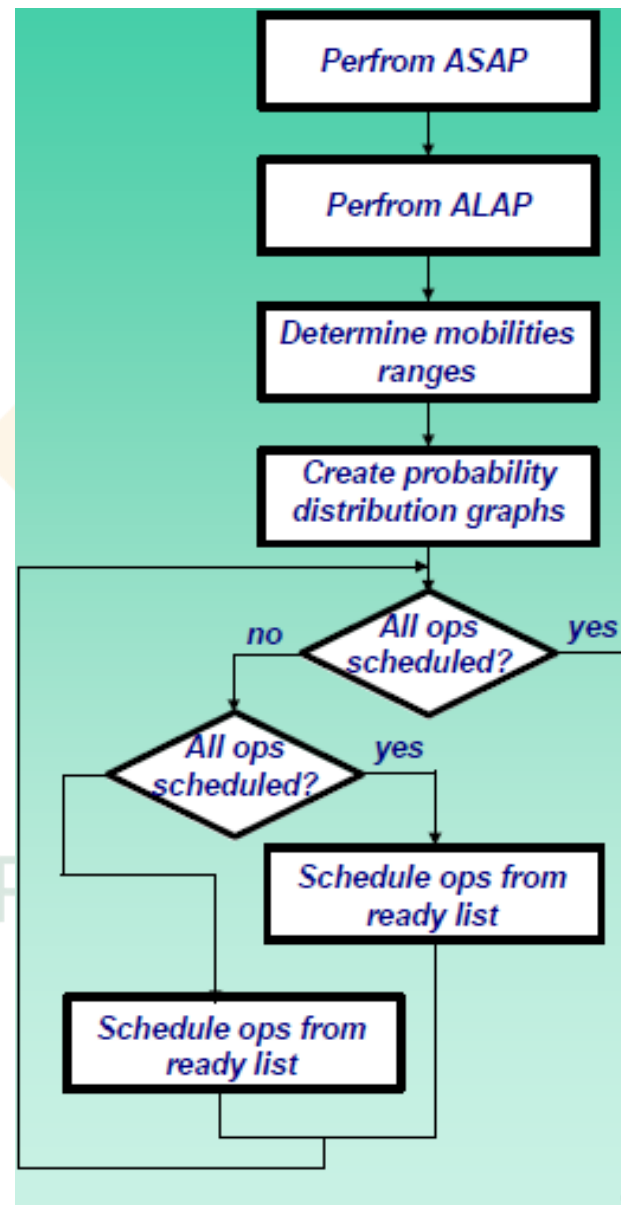


(a) ASAP

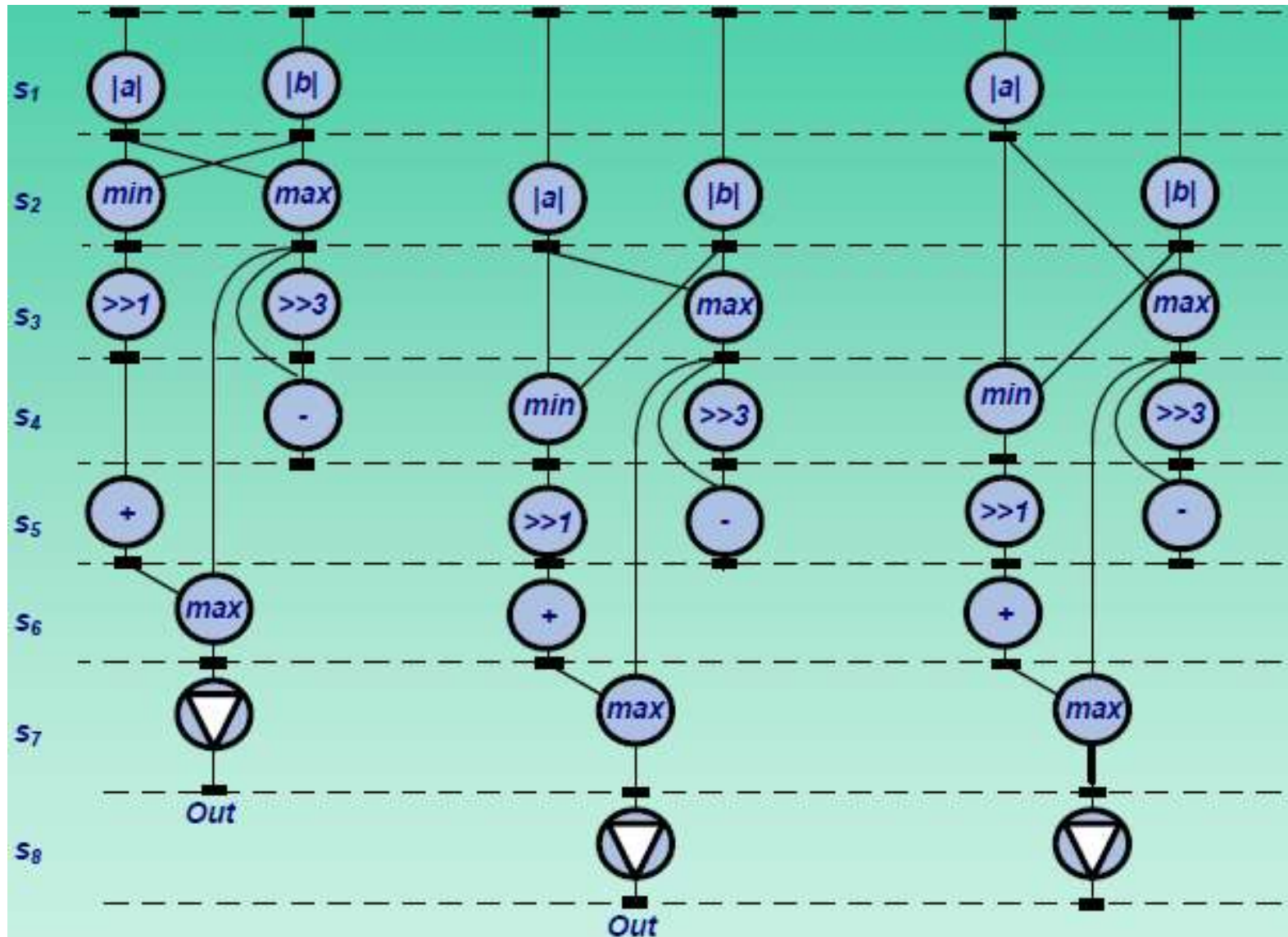
(b) ALAP

(c) Ready list
with mobilities(d) RC
schedule

Time-constrained scheduling



TC schedule for SRA algorithm

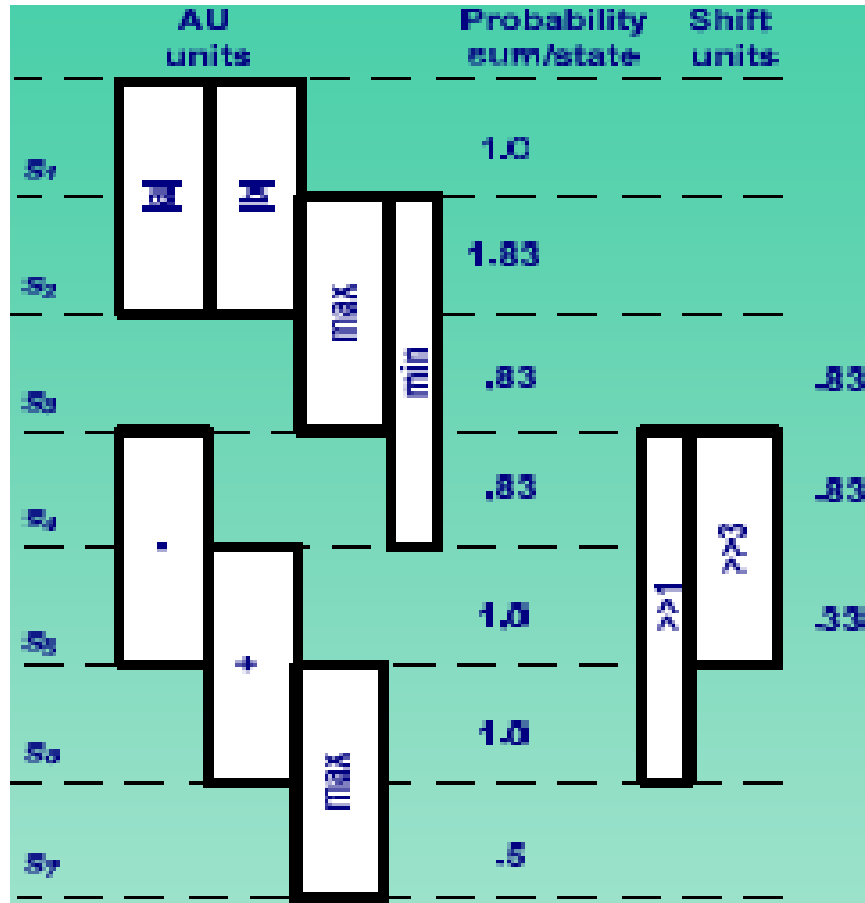


(a) ASAP

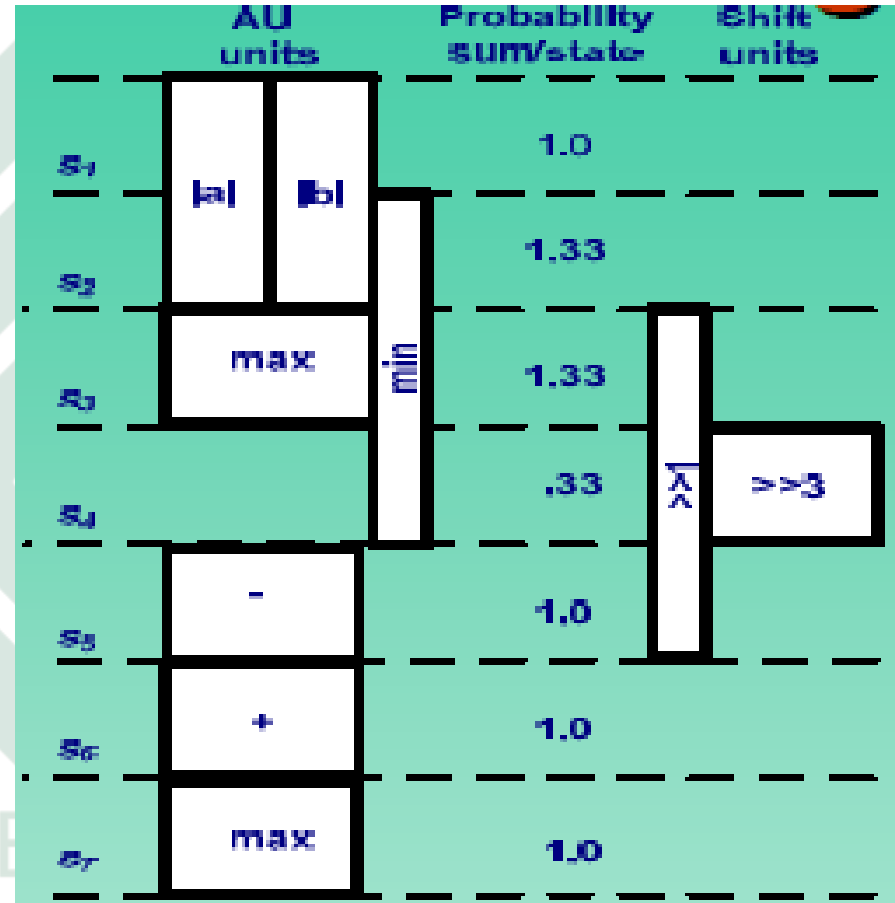
(b) ALAP

(c) TC schedule

Probability distribution graph before, during and after TC scheduling



(a) Initial probability distribution graph



(b) Distribution graph after max, + and - were scheduled

Probability distribution graph before, during and after TC scheduling

	AU units	Probability sum/state	Shift units	
s_1	[a] [b]	1.0		
s_2		1.0		
s_3	max	1.0		
s_4	min	1.0	>>3	1.0
s_5	-	1.0	>>1	1.0
s_6	+	1.0		
s_7	max	1.0		

(c) Distribution graph after max, + and -, >>3 and >>1 were scheduled

	AU units	Probability sum/state	Shift units	
s_1	[a]	1.0		
s_2	[b]	1.0		
s_3	max	1.0		
s_4	min	1.0	>>3	1.0
s_5	-	1.0	>>1	1.0
s_6	+	1.0		
s_7	max	1.0		

(d) Distribution graph for final scheduled

Chapter summary

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